

1. What is Web 3.0, and how is it different from Web 1.0 and Web 2.0?

- Web 3.0 is the advancement of Web 2.0. Web 3.0 features AI implementation to personalize things like ads, interconnection of multiple applications, and more cloud-based reliance.

2. What are the functional differences between the front end of a web application and its back end?

- The front end of a web application is what the user sees. The user interacts with the HTML, CSS, and JavaScript to receive and send information.
- The back end of a web application is what the user does not see. This is the server side which stores and fetches data to the user.

3. In your own words, explain the process that takes place from when you type a URL into the address bar in your browser until you finally view the page you have requested. Include the HTTP request–response cycle in your answer and provide an example of the HTTP request and response messages.

- When the user types a URL into the address bar, the browser uses this address to make a connection with the server. It sends the server an HTTP request, like GET / HTTP/1.1, and the server sends back an HTTP response, like HTTP/1.1 200 OK. In this response, if successful, the webpage's HTML content which is interpreted by the browser, so the user can see the text and images in the requested page.